Beyond the Syllabus — Exploration 01

When Boundaries Move

From the Fundamental Theorem of Calculus to the Conservation of Area

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The question behind the question.

Question 32 of the 2026 HSC Mathematics Advanced examination considers the function

$$g(x) = \int_{x^3}^{1} \sqrt{\tan t}\, dt, \qquad 0 \leq x \leq 1.$$

If F is an antiderivative of $\sqrt{\tan t}$, the Fundamental Theorem of Calculus gives $g(x) = F(1) - F(x^3)$. Differentiating with the chain rule,

$$g'(x) = -3x^2\sqrt{\tan(x^3)}.$$

There is nothing especially lengthy about this calculation. The more interesting question is what it *means*. How can we know how quickly an area changes without first finding the area itself?

Since $\sqrt{\tan t} \geq 0$ on $0 \leq t \leq 1$, the integral represents the area beneath the curve between the moving lower boundary $t = x^3$ and the fixed boundary $t = 1$. As x increases, the lower boundary moves right, removing area. The derivative tells us the instantaneous rate of that loss: the minus sign indicates removal, $3x^2$ is the boundary's velocity with respect to x, and $\sqrt{\tan(x^3)}$ is the height of the curve where the movement occurs. Although the curve itself rises as t increases, the remaining area $g(x)$ falls. A rising integrand need not produce a rising integral function.

Why do height and speed appear together?

Imagine moving the lower boundary of a region beneath a continuous, non-negative curve $y = f(t)$ from u to $u + h$. The lost region is a thin strip. If h is small, its area is close to that of a rectangle of width h and height $f(u)$, so the area lost is approximately $f(u)h$. The curve need not be a straight line: continuity ensures that its height changes very little across a sufficiently narrow strip. More precisely,

$$\lim_{h\to 0} \frac{1}{h} \int_{u}^{u+h} f(t)\, dt = f(u).$$

The height $f(u)$ is therefore the exact instantaneous area change *per unit distance* moved by the boundary. If the boundary's position is itself a function $u = a(x)$, we must also account for how quickly it travels. The Fundamental Theorem of Calculus and the chain rule give

$$\frac{d}{dx} \int_{a(x)}^{c} f(t)\, dt = -f(a(x))a'(x).$$

This recovers the meaning of all three factors in the HSC derivative. We have used the behaviour of the integrand *at the boundary*, rather than an explicit evaluation of the entire area.

Two moving boundaries.

Now imagine that the right boundary can move as well. As the left boundary removes area, the right one can add it. Could these changes balance exactly? To investigate, begin with $f(t) = t$ on $t \geq 0$, where the area between two boundaries $0 \leq a < b$ is an ordinary trapezoid:

$$A = \tfrac{1}{2}(a+b)(b-a) = \tfrac{1}{2}(b^2 - a^2).$$

Thus a fixed area $K > 0$ does not require either boundary to remain fixed; it requires only that $b^2 - a^2 = 2K$. When $a = 1$ and $b = 3$, the area is 4. If the left boundary moves to $a = 2$, the right boundary must move to $b = \sqrt{12}$ to preserve those four square units.

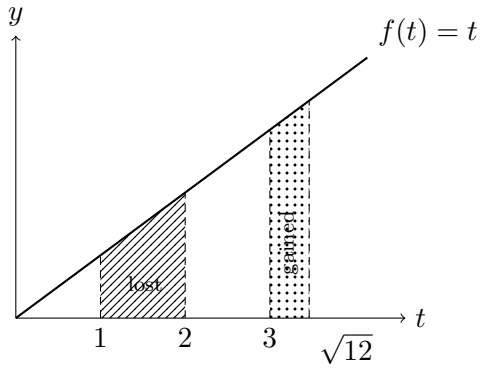


Figure 1. The hatched region lost between 1 and 2 equals the dotted region gained between 3 and $\sqrt{12}$; each has area 1.5.

The boundaries have both moved to the right, but by different distances: the left boundary travels 1 unit while the right travels only $\sqrt{12} - 3 \approx 0.464$ units. This makes geometric sense. The curve is taller on the right, so a narrower strip can replace the area of a wider strip on the left. *Conserving area does not require conserving width.*

There is also a relationship between the boundaries' instantaneous velocities. If a and b depend differentiably on a parameter x, then differentiating $b(x)^2 - a(x)^2 = 2K$ gives

$$b(x)b'(x) = a(x)a'(x).$$

Each side is a *height multiplied by a velocity.* When $b > 0$, we may equivalently regard b as a function of a and write $db/da = a/b$. This ratio can change as the boundaries move: it is not a fixed ratio of the total distances travelled.

Beyond straight lines.

The conclusion survives when the straight line is replaced by a curved graph. For a continuous function f and differentiable boundaries $a(x) \leq b(x)$, write

$$A(x) = \int_{a(x)}^{b(x)} f(t)\,dt = F(b(x)) - F(a(x)), \qquad F' = f.$$

The chain rule now gives the moving-boundary form of *Leibniz's rule*:

$$A'(x) = f(b(x))b'(x) - f(a(x))a'(x).$$

Conserving the area.

For a positive integrand with both boundaries moving to the right, the first term measures the rate at which area arrives and the second the rate at which it leaves. On an interval, A remains constant exactly when these rates agree throughout:

$$\boxed{f(b(x))b'(x) = f(a(x))a'(x).}$$

The striking point is not the extra formula. It is that *an unchanged area can be the result of two continuous, precisely balanced changes.*

Return now to the HSC question. Its upper boundary is fixed, so it cannot add area to replace what the lower boundary removes. Indeed, for $0 < x < 1$, both $\sqrt{\tan(x^3)}$ and $3x^2$ are positive, and hence $g'(x) < 0$. The examination derivative is the one-boundary case of the same general principle.

Taking the investigation into a classroom.

This is an opportunity to teach the meaning of the Fundamental Theorem of Calculus rather than just its procedure. After students differentiate the HSC integral, ask why the answer is negative when $\sqrt{\tan t}$ is increasing. Have them identify the moving boundary, predict what happens to the shaded area and account for each factor in $g'(x)$. This reveals a common misconception: students may confuse the slope or height of the integrand with the rate of change of an accumulated area.

Then pose a problem that does not initially require a new calculus rule: "The area under $y = t$ between 1 and 3 is 4. If the left boundary moves to 2, where must the right boundary go to keep the area at 4?" Students can solve this using only the trapezoid area formula, compare the lost and gained regions and explain why the boundaries travel different distances. Once the geometry is understood, differentiation of the constant-area equation leads naturally to the general conservation condition. A teacher could use this as a short enrichment discussion or as a follow-up to a lesson on the Fundamental Theorem of Calculus.

Explore the interactive.

The companion investigation allows readers to drag both boundaries under $f(t) = t$ or $f(t) = t^2$, then activate *Conserve Area* to see one boundary respond to the other. Its HSC Connection tab returns to Question 32 and displays both the changing shaded area and the derivative. **Explore the interactive online**, or scan the QR code.

Source. Inspired by Question 32, 2026 HSC Mathematics Advanced examination, p. 37, NSW Education Standards Authority (NESA). Official examination paper. The interpretation, extension and diagram in this article are independently developed.